Explainable Artificial Intelligence for Automated ECG Diagnosis and Patient Interpretation

Krishna Unnam[1], Manhitha Mamillapalli[1], Sanjay Lakshmanan[1], Stephen Barretto[1], Medha Vemparala[1], Dominic Seaney[2], Dr. Paul Nguyen MD[3]

[1]Department of Bioengineering, [2]Department of Statistics, [3]Sudden Cardiac-death Awareness Research Foundation

University of Illinois at Urbana Champaign, Sudden Cardiac-death Awareness Research Foundation

Introduction

- **Electrocardiograms (ECGs)** are widely used diagnostic tools that **capture cardiac electrical activity across cycles**, with the QRS complex representing ventricular electrical activation.

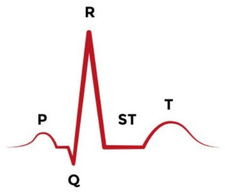


***Figure 1**. QRS complex of an ECG wave showing the change in ventricular electrical activation*

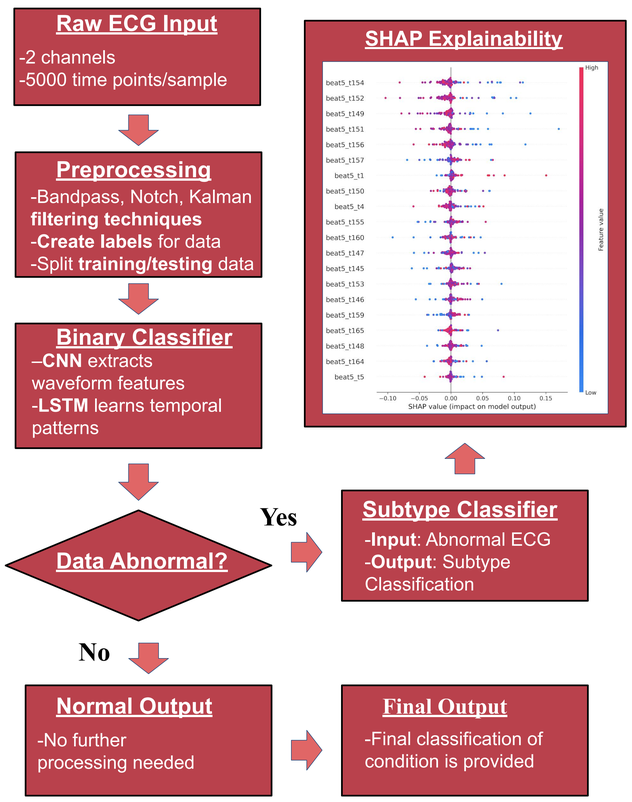
Current Problems

- Automated systems present results that require expertise to interpret, **limiting patient comprehension.**
- The increase in portable ECG devices has increased t**he volume of cardiac data** but also **introduced challenges to interpret data** due to **noise from patient movement**

Hypothesis

A **CNN–LSTM–based ECG analysis pipeline** can **analyze cardiac rhythms** by capturing **waveform morphology** and **temporal dependencies across consecutive heartbeats**. Additionally, SHAP-based explainability can identify **clinically meaningful ECG regions** and **irregular rhythm patterns** that contribute to model predictions.

Methodology



Results

***Figure 2**. Binary Classification Accuracy Performance of Model (Accuracy : 0.99)*

Class	Precision	Recall	F1-Score	Support
Normal	1.00	0.99	0.99	18085
Abnormal	0.94	0.98	0.96	2165
Accuracy	n/a	n/a	0.99	20250
Macro Avg	0.97	0.99	0.98	20250
Weighted Avg	0.99	0.99	0.99	20250

***Figure 3.** Subtype Classification Accuracy Performance of Model*

Class	Precision	Recall	F1-Score	Support
Supraventricular	0.75	0.97	0.85	556
Ventricular	0.98	0.97	0.97	1443
Fusion	0.53	0.97	0.69	160
Unknown	0.24	0.97	0.35	6
Accuracy	n/a	n/a	0.98	20250
Macro Avg	0.70	0.91	0.77	20250
Weighted Avg	0.99	0.98	0.98	20250

***Figure 4.** ROC/Precision Recall Curve of Effectivness of Binary Classification*

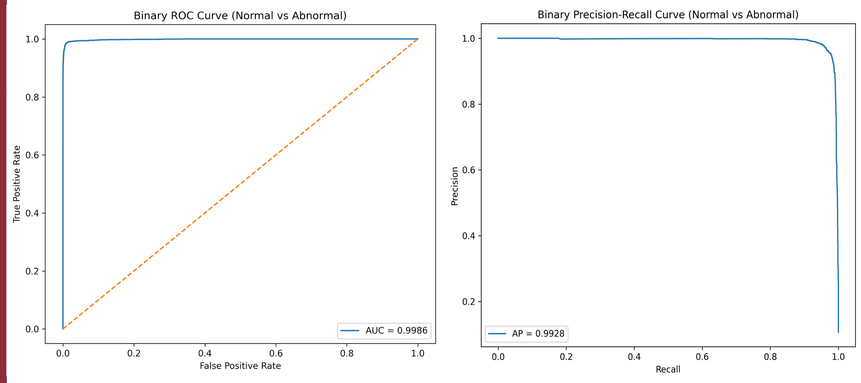
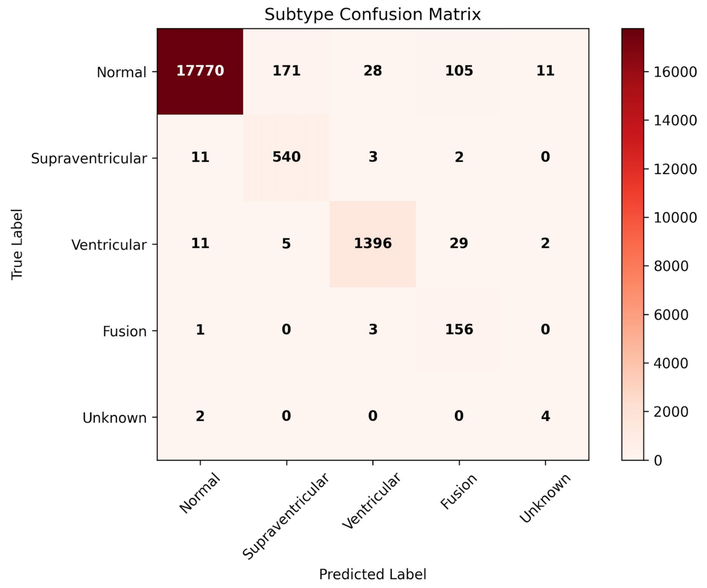


***Figure 5.** Confusion Matrix of Subtype Classification Model*



Conclusion

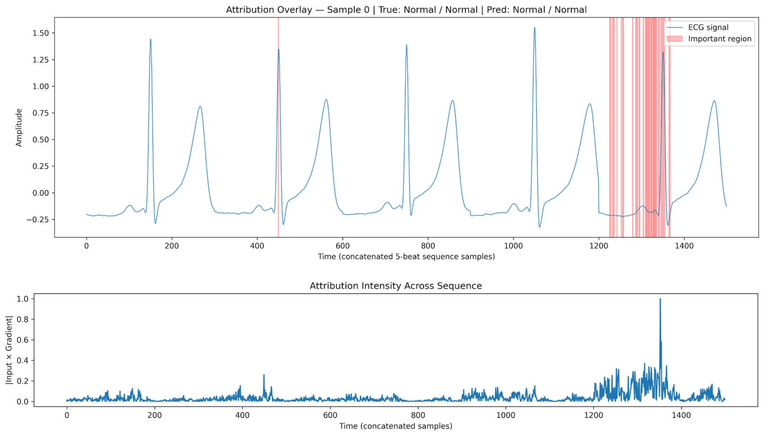
End-to-End ECG Analysis

This study presents a **comprehensive ECG analysis** that integrates **signal denoising**, **deep learning classification**, and p**atient- accessible interpretation** into a simple framework.

Model Performance and Classification Limitations

- Model demonstrates solid performance in binary classification achieving an **overall accuracy of 85%.**
- Very effective at **identifying abnormal signals** with **precision (0.99)** and strong **F1-score (0.90)**
- **Less balanced for normal signals**, where **lower precision (0.59)** suggests the model may label normal cases as **abnormal**.

***Figure 6.** Comprehensive SHAP analysis of all ECG data*



Interpretability and Patient Accessibility

- **Segment level explainability** module links predictions to **specific parts of the PQRST waveform** as seen in Figure 6
- Emphasizes **interpretability and accessibility**
- Helps users understand **how decision are made**

Future Directions

- Need for **accurate** and **interpretable** analysis systems is **increasingly important.**
- **Further improvements** for **underrepresented classes** for consistent performance **across all categories.**
- **Integration of a conversational chatbot** translates the **technical ECG outputs to understandable responses,** increasing patient accessibility and data understanding.

References

1. Hossain, B., Bashar, S. K., Walkey, A. J., McManus, D. D., & Chon, K. H. (2019). An Accurate QRS complex and P wave Detection in ECG Signals using Complete Ensemble Empirical Mode Decomposition Approach. IEEE access: practical innovations, open solutions, 7, 128869–128880. https://doi.org/10.1109/access.2019.2939943
2. Wagner, P., Strodthoff, N., Bousseljot, R., Samek, W., & Schaeffter, T. (2022). PTB-XL, a large publicly available electrocardiography dataset (version 1.0.3). PhysioNet. RRID:SCR_007345. https://doi.org/10.13026/kfzx-aw45
3. Goldberger, A., Amaral, L., Glass, L., Hausdorff, J., Ivanov, P. C., Mark, R., ... & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation [Online]. 101 (23), pp. e215–e220. RRID:SCR_007345.